CSC4197 — RESEARCH METHODOLOGY

PRESENTATION TECHNIQUES, VIVA VOCE & MINI-CONFERENCE

2 WHY PRESENTATION SKILLS DECIDE OUTCOMES

- The same research can sound groundbreaking or boring — **delivery is part of the contribution**
- Where you'll need it: thesis defense, conference talks, job interviews, demo days, funding pitches
- Today: slide craft → talk delivery → viva tactics → then **you present**

3 SLIDE DESIGN PRINCIPLES

- **One idea per slide** — the slide title should state the idea ("LLM feedback raised scores by 11%"), not the topic ("Results")
- **Minimal text:** ≤ 6 lines, ≥ 24 pt; slides are visual aids, not your script
- **Figures > tables > text** — redraw paper figures for slides (bigger fonts, fewer series)
- High contrast, consistent template, no decorative clutter/animations
- Numbered slides (helps Q&A: "back to slide 7")
- Budget: **~1 slide per minute**, hard stop at the limit

4 STRUCTURING A 10-MINUTE RESEARCH TALK

Minutes	Content
0–1	Hook + problem (why should the audience care?)
1–2	Gap + research question
2–4	Method (one architecture/flow diagram beats 5 text slides)
4–7	Results (2–3 key figures, stated plainly)
7–8.5	Discussion + limitations (own them before examiners do)
8.5–10	Takeaway slide + "Thank you — questions?"

Storytelling arc: situation → complication → resolution. The audience should be able to repeat your one-sentence takeaway afterward.

5 DELIVERY TECHNIQUES

- **Rehearse aloud 3×** — once alone, once timed, once before friends; rehearsal is the #1 predictor of talk quality
- Open strong: a question, statistic, or 15-second scenario — never "umm, so, our topic is..."
- Eye contact with sections of the room; don't read slides; face the audience, not the screen
- Pace: pause after key claims; nerves accelerate speech — breathe
- Demos: pre-recorded video > live demo (the demo gods are cruel)
- Team talks: clean handoffs, everyone presents, everyone can answer anything

6 THEORY: WHAT IS A VIVA VOCE?

- Oral examination defending your work: presentation + sustained questioning
- Examiners probe: **understanding** (is it truly yours?), **soundness** (do methods hold?), **awareness** (limitations, alternatives, literature)
- It is a **scholarly conversation**, not an interrogation — examiners *want* to find competence
- Typical verdicts: pass / pass with minor or major corrections / re-defense

7 THE VIVA QUESTION BANK (PREPARE THESE!)

- **Always asked, in some form:**
- Summarize your research in 2 minutes.
- Why this problem? Why is it important?
- Why this method and not [obvious alternative]?
- How did you decide the sample/dataset? Why that size?
- What are the limitations / threats to validity?
- What would you do differently with more time?
- What is your **original contribution**?
- How does your work compare with [paper X]?
- Are your results generalizable? To whom?
- What are the practical/ethical implications?
- Write your answers **before** the defense. No examiner question should be a surprise.

8 VIVA FACING TECHNIQUES

- **Listen fully; pause 2–3 seconds** before answering — composure reads as competence
- Clarify when unsure: "If I understand correctly, you're asking whether..."
- **Structure answers:** direct answer → evidence from thesis ("as shown in Table 5.2") → acknowledge boundary
- Honest "I don't know": "*That's beyond what I investigated, but I would expect... and testing it would be valuable future work*" — far better than bluffing
- Defend without defensiveness: concede valid criticism gracefully; that earns marks
- Body language: upright, hands calm, address the *whole* panel, smile when natural
- Know every number in your own tables — examiners love picking one and asking "why?"

9 CASE STUDY: TWO VIVAS, SAME THESIS

- **Student A** opens slides reading verbatim, answers Q1 with a 6-minute ramble, claims their accuracy "proves" the model is best, gets defensive when sampling is challenged → major corrections.
- **Student B** delivers a tight 10-minute story, pauses before answers, says "our convenience sample limits generalization to one university — a multi-site study is the natural next step," cites their own Table 4.3 from memory → pass, minor corrections.
- **Same research.** Discuss: list five behavioral differences and rank their impact.

10 COURSE WRAP-UP: YOU ARE NOW RESEARCHERS

- **The full pipeline you've practiced:**
  - problem → gap → RQ/hypothesis → design → sampling → data → paper (LaTeX/Word + Zotero) → ethics & AI responsibility → dissemination → defense ✅
- **Where to go next:**
  - Polish your mini-paper → submit to a student track/national conference
  - Thesis next year: you have the entire toolkit
  - Grad school: SOP + research proposal build directly on these skills (talk to me!)
  - Keep your ORCID, Scholar profile, and Zotero library alive



Thank you!
Go discover something.